

EDUCATION

Bachelor of Engineering in Computer Engineering (Artificial Intelligence)

Azad University, North Tehran Branch

Sep 2016 – 2021

Tehran, Iran

- Relevant coursework: Artificial Intelligence, Computer Vision, Image Processing, Data Structures and Algorithms, Software Engineering, Database Systems, Computer Networks, Operating Systems, Discrete Mathematics, Probability and Statistics.

SKILLS

- **Programming:** Python, Bash, JavaScript, HTML/CSS.
- **Infrastructure & Systems:** VMware ESXi, Ubuntu, Windows Server, TrueNAS, ZFS, Veeam, MikroTik, TCP/IP, DNS, DHCP, FreePBX.
- **DevOps & Cloud:** Docker, Docker Compose, Kubernetes (in progress), Git, GitLab CI (in progress), GitHub Actions.
- **AI & Data Science:** OpenCV, NumPy, Pandas, scikit-learn, pyzkl, Matplotlib, Image Processing, Feature Extraction, SVM, KNN.
- **Languages:** Persian (Native); English (Professional working proficiency, IELTS in preparation).

RESEARCH INTERESTS

Applied AI and Computer Vision

Image processing, object detection, and pattern recognition in scientific and real-world domains.

Cloud-Native Systems Architecture

Distributed, containerized infrastructure and high-availability workloads at scale.

MLOps and Intelligent Automation

Production ML deployment, containerization, CI/CD, orchestration, and predictive operations.

Software Reliability and Performance

Fault tolerance, observability, deployment strategies, and empirical systems analysis.

WORK EXPERIENCE

Systems and DevOps Engineer

Khademan Emam Zaman (shamim313.com)

Jan 2025 – Present

Mashhad, Iran

- Administered a multi-node VMware ESXi environment hosting approximately 10 production VMs for a nationally distributed platform serving 100,000+ registered users.
- Containerized 5+ production services with Docker, reducing deployment time by approximately 10% and limiting configuration drift.
- Automated backup and maintenance workflows using Veeam backup tools and Bash, recovering approximately 3–4 hours per week and reducing manual errors.
- Maintained zero-downtime operation for 6+ months on a FreePBX VoIP system supporting 20+ staff extensions.
- Managed firewall rules, DHCP, access control, and TrueNAS ZFS storage with automated snapshots for 1–10 TB of data.

DevOps Engineer

Part Software Group

Apr 2024 – Jan 2025

Mashhad, Iran

- Managed 200+ virtual machines across development, staging, QA, and production environments.
- Standardized deployment workflows using the company's in-house Docker container platform (Kakado), and supported OS hardening, patching, monitoring, and weekly releases across 4+ environments.

Mandatory Military Service

Islamic Republic of Iran Armed Forces

Apr 2022 – 2024

Iran

- Completed 24-month compulsory national service.

RESEARCH PROJECTS

Automated Detection and Classification of Solar System Planets Using Image Processing 2020–2021

Bachelor's Internship Project, Azad University, North Tehran Branch

- Designed a computer vision pipeline for detecting and classifying planets in astronomical imagery from publicly available NASA and ESA archives.
- Applied Gaussian and median filtering, histogram equalization, CLAHE, background subtraction, Canny edge detection, and OpenCV contour analysis.
- Extracted morphological, photometric, color-histogram, and GLCM texture features, then evaluated SVM and KNN classifiers for multi-class planetary identification.
- **Technologies:** Python, OpenCV, NumPy, scikit-learn, Matplotlib.

COURSES AND CERTIFICATES

- **Crash Course on Python** — Google / Coursera — [Completed](#)
- **Using Python to Interact with the Operating System** — Google / Coursera — In progress
- **Kubernetes Administration (CKA path)** — Self-directed lab environment — In progress

VOLUNTEER AND EXTRACURRICULAR ACTIVITIES

- IT Training Volunteer — computer repair and IT literacy training for non-technical users.
- Volunteer IT & ICDL Instructor — Designed and delivered training sessions on core ICDL modules (including file management, spreadsheets, and online security) for non-technical adult learners.
- Graphic Design Volunteer — cultural and administrative materials for Azad University Cultural Affairs Department.